

Antonio Carmagnani

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Dear Hiring Team at NIRAS,

NIRAS sits where I have been trying to take my work, at the meeting point of sustainable buildings, environmental performance, and data. I trained as an architect in Brazil, but what held me was less the drawing than the performance behind it, how a building uses energy and daylight, what it is made of, and whether those choices can be measured and improved. Following that took me to an M.Arch. in Architecture Computation at The Royal Danish Academy, focused on data-driven approaches to sustainability. I am reaching out because your Building Sustainability, Indoor Climate and Energy practice, and NIRAS's wider push to bring data and digital tools into how buildings perform, is the work I want to do.

Even in architecture practice, the part I gravitated to was environmental. At Isay Weinfeld and earlier studios I ran daylight and solar-exposure simulations with Ladybug in Grasshopper to drive massing and orientation, quantifying light across building layouts rather than guessing at it. At Itaú-Unibanco I assessed the environmental and energy performance of buildings the bank was acquiring or leasing, reviewing consumption data, efficiency ratings, and certifications. Reading a building's performance from data and turning it into a decision is the thread that connects everything I have done.

That work rests on a computational and sustainability base. I collaborated with a Copenhagen LCA startup to build a Revit toolkit that computes embodied carbon from GWP values across life-cycle stages, and a semester project on a blue-biomass material system, developed with LCA metrics and parametric simulation, became the material foundation for the school's installation at the 2025 Venice Architecture Biennale. At Ramboll I worked on real estate ESG due diligence centered on building energy and decarbonization under EU building performance regulations. Python, QGIS, and Revit are working tools for me, not references on a page.

NIRAS pairs deep building engineering with a real investment in data and digitalisation, and it does that from Copenhagen, where I am already based. Bringing computational and data methods to how buildings are made to perform is the direction I want to grow in, and NIRAS is where I would most want to do it.

All the best,

Antonio Carmagnani